

TFDV PRODUCTION USE-CASE



What TFDV is *for*

TFDV is built primarily for **batch data validation**, in two core use cases:

1. Before training (data validation & schema inference)

- Infer schema from training data.
- Detect anomalies in training/eval data.
- Identify type issues, missing values, drift, etc.

2. Post-deployment drift detection (batch monitoring)

- Once per hour/day, process *logged inference data* (from S3) to detect drift.
- Compare against training baseline.
- Generate anomaly reports.

✅ This is **offline** monitoring. Data is collected, stored (usually in S3), then processed in **batch** using TFDV in a scheduled job or pipeline.



What TFDV is not designed for:

- Real-time or low-latency validation per request
- Inference-time input validation
- Streaming data monitoring

TFDV doesn't work on a single data point or row — it needs **distributions over many examples**.



So how do you monitor model *in production*?

You shift to **two modes of monitoring**:



1. Real-time monitoring (data capture + logging)

- Enable **data capture** on your **SageMaker endpoint** (CaptureConfig).
- It stores inputs & outputs in S3.
- You can then run TFDV **periodically (e.g. daily)** over that captured data.

✓ Now you have **real user data**, and can run TFDV (batch) to compare this against your training baseline.

```
python
from sagemaker.model_monitor import DataCaptureConfig

data_capture_config = DataCaptureConfig(
    enable_capture=True,
    sampling_percentage=100,
    destination_s3_uri='s3://my-bucket/capture/',
    capture_options=['Input', 'Output']
)
```

✓ 2. Lightweight checks at inference time (optional)

For real-time safety, you can also:

- Add basic **input validators** in your inference code (e.g. NaN checker, value range checker).
- Use libraries like pydantic, cerberus, or your own logic to reject invalid inputs.
- Log failed requests.

⚠ These aren't as powerful as TFDV (no drift, schema inference, etc.), but **give you safety against malformed requests**.

📄 Typical Architecture

